Donchian Channel

Indicators & Analysis / Trend | beginner | 4 min read | DeepCharts Help Center | August 24, 2026

The Donchian Channel plots two lines around price: the highest high and the lowest low of a chosen number of past bars. The result is a channel that contains all recent price action, making breakouts and consolidations unmistakable.

It is one of the simplest indicators to understand — there is exactly one parameter — which makes it a good first channel indicator before moving on to volatility-based bands.

What it is

The channel answers the question "where are the recent extremes, and is price breaking beyond them?" It consists of two elements:

- **Upper Line** — the highest high of the lookback period.
- **Lower Line** — the lowest low of the lookback period.

When price touches or exceeds the upper line, it is making a new high for the period; the same logic applies to the lower line for new lows. The vertical distance between the lines is a direct read of recent range: a widening channel means volatility expansion, a narrow channel means compression.

When to use it

- To trade breakouts: a close beyond the channel means price has exceeded every high or low in the lookback window.
- To read trend direction: in an uptrend the upper line steps upward repeatedly while the lower line trails behind.
- To spot volatility expansion and contraction from the channel width.
- To locate support and resistance: the channel lines mark the period's extremes, which often act as reaction levels.

Quick start

1. Open a chart and click the bar-chart icon in the top-left corner to open the **Indicators** panel.
2. Click **Indicators** to open the full **Indicator List**.
3. Search for "Donchian Channel" and click **+** to add it.
4. Click the gear icon next to the indicator to open its settings and set **Length** to the lookback you want to trade against — a shorter length hugs price for short-term signals, a longer one defines the bigger structure.

[SCREENSHOT PLACEHOLDER] A price chart with the Donchian Channel plotted — upper and lower lines stepping around the candles, with a recent breakout candle closing above the upper line `dc-en-donchian-channel-01.png`

How to read it

- **Breakout above the Upper Line**: price has printed a new high for the entire lookback period — bullish momentum. A breakout below the Lower Line is the bearish mirror.
- **Stair-stepping lines**: an upper line that keeps ratcheting higher while the lower line rises behind it is trend behavior; use the lower line as a trailing reference.
- **Flat, narrow channel**: price has not made a new extreme in the whole window — consolidation. Compressed channels often precede expansions, so a squeeze is a reason to watch for the next breakout.

• **Channel width**: widening = volatility expanding; narrowing = volatility contracting. Compare with ATR if you want the same information as a single number.

Settings reference

Parameters

Setting	What it does
Length	How many past bars are used to calculate the highest high (Upper Line) and lowest low (Lower Line). Increasing the length widens the channel and slows its reactivity; decreasing it makes the channel more sensitive to recent price movement.

Subgraphs

Setting	What it does
Donchian_Channel	Controls the visual styling of the indicator: line colors, subgraph style, auto-coloring, line style, line width and a custom short-name label.
+DI (Upper Line)	The subgraph representing the highest high over the selected period.
-DI (Lower Line)	The subgraph representing the lowest low over the selected period.

> **Note:** The upper and lower lines appear in the settings dialog under the subgraph names **+DI** and **-DI**. They are the Donchian upper and lower lines, not the directional-movement lines from ADX.

Each subgraph also exposes display options: **Name Label**, **Value Label**, **Name Background**, **Value Background**, **Include on Auto Center**, and **Chart color for marker** — these control how the line is labeled and whether it influences the chart's auto-centering.

[SCREENSHOT PLACEHOLDER] The Donchian Channel settings dialog showing the Length parameter and the subgraph rows for Donchian_Channel, +DI (Upper Line) and -DI (Lower Line) with their display options
`dc-en-donchian-channel-02.png`

Tips and common mistakes

• **Match Length to your holding period.** A day trader reading a 5-minute chart and a swing trader reading a daily chart should not use the same lookback; the channel is only meaningful relative to the horizon you trade.

• **A touch is not a breakout.** Price tagging the upper line only means it equaled the period high; many traders wait for a close beyond the channel to filter noise.

• **Expect whipsaws in ranges.** In sideways markets price repeatedly pokes both extremes; combine the channel with a trend-strength filter such as ADX before trading breakouts.

• **Don't confuse it with volatility bands.** Bollinger Bands and the Keltner Channel are built from averages and volatility; the Donchian Channel is pure price extremes, so it behaves differently around quiet periods.

Related articles

• Keltner Channel
• Bollinger Bands
• Average True Range (ATR)
• Average Directional Index (ADX)

- Different Types of Input Data for Indicators
- How to Change the Layout of Indicators